# POSTNIKOV TRIPLES, THEIR MAPS, AND THE SPHERES: AN OVERVIEW OF THE ARGUMENT

This note describes, at a high level, the argument formalised in `sag/pi3-s2-postnikov` by the modules `TripleMaps`, `TripleMapsEquiv`, `TripleComponents`, `TripleHomotopy`, `Pi3TripleUniversal` and `Pi4TripleUniversal`. The detailed informal proof, with a declaration-by-declaration correspondence, is `doc/postnikov-morphisms.pdf`.

**Attribution.** The strategy is David Wärn's, as communicated by the user: classify pointed simply connected 3-types *and their arbitrary pointed maps* by Postnikov data, identify the universal objects of the spheres on the side of the data, and treat $\pi_4(S^3)$ by embedding pointed 2-connected 4-types into pointed simply connected 3-types by looping. The classification of objects is Hart–Milner's (Owen Milner's equivalence). We do not know Owen Milner's unpublished treatment of maps; nothing here claims to reproduce it.

## 1. The plan

To compute $\pi_3(S^2)$ it suffices to know a pointed simply connected 3-type $P$ with a class $s \in [S^2, P]_*$ such that

$$s^* : [P, X]_* \to [S^2, X]_*$$

is a bijection for every pointed simply connected 3-type $X$: then $P$ is the 3-truncation of $S^2$, and $\pi_3(S^2) = \pi_3(P)$ by a Yoneda argument. Similarly, $\pi_4(S^3)$ is computed from a pointed 2-connected 4-type with the analogous universal property among such types.

Wärn's idea is to make the left-hand side of this bijection *algebraic*. If pointed simply connected 3-types and their maps are described by algebraic data, then $[P, X]_*$ becomes a set of classes of algebraic morphisms, computable from the cohomology of Eilenberg–MacLane spaces, while $[S^2, X]_*$ is computed from the sphere. The argument has three steps:

(1) classify the objects (done earlier: `PostnikovClassification`);
(2) classify the arbitrary pointed maps (new: Theorem 1);
(3) compute the sphere's universal property on the algebraic side (new: Sections 5 and 6), and recover the homotopy group.

## 2. Objects

A *Postnikov triple* is $t = (A, B, k)$ with abelian groups $A, B$ and a pointed map $k : K(A, 2) \to_* K(B, 4)$. Its realisation is the fibre

$$\mathrm{Fib}k = \textstyle\sum_{x:K(A,2)} (k\, x = \mathrm{pt}),$$

a pointed simply connected 3-type with $\pi_2 = A$ and $\pi_3 = B$. The map

$$\mathsf{realise} : \mathsf{Triples} \to \mathsf{SC3}$$

is an equivalence of types (`PostnikovClassification.classification`). Both types live in the next universe and are not sets; they are 2-types.

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## 3. Maps

For triples $t = (A, B, k)$ and $t' = (A', B', k')$ define

$$\mathsf{Mor}(t, t') = \sum_{a:\mathrm{Hom}(A,A')} \; \sum_{b:\mathrm{Hom}(B,B')} \big(K(a,2)\,;\, k' = k\,;\, K(b,4)\big).$$

Here $f\,;\,g$ means "$f$, then $g$", and the last component is a *path of pointed maps*: a homotopy together with its coherence at the base point. Such a morphism determines a map of fibres, $(x, q) \mapsto (K(a,2)\,x,\ H_x \cdot K(b,4)(q))$. This is the canonical construction `FibMap.fm` already present in the library.

**Theorem 1** (`TripleMapsEquiv.Maps2.realise-map-equiv`)**.** *For all triples $t, t'$, the canonical map*

$$\mathsf{Mor}(t, t') \longrightarrow \mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$$

*is an equivalence of types. There is no hypothesis on the groups or on $k, k'$, and no sphere axiom.*

Both sides are 1-types, not sets in general. The homotopy $H$ is genuine data. For $t = (A, 0, 0)$ and $t' = (0, C, 0)$ the coefficients carry no information at all, and both sides are $\mathrm{Map}_*(K(A,2), K(C,3))$, whose components form $H^3(K(A,2); C)$.

**Idea of the proof.** A pointed map $\mathrm{Fib}k \to \mathrm{Fib}k'$ is a map to $K(A', 2)$ together with a pointed null-homotopy of its composite with $k'$.

- *The base.* The projection $\mathrm{Fib}k \to K(A,2)$ has 2-connected fibres, and $K(A',2)$ is a 2-type. So maps into $K(A',2)$ factor uniquely through $K(A,2)$, and $\mathrm{Map}_*(K(A,2), K(A',2)) \simeq \mathrm{Hom}(A, A')$.
- *The key lemma.* For $c : K(A,2) \to_* K(B',4)$, pointed null-homotopies of $\mathrm{Fib}k \to K(A,2) \xrightarrow{c} K(B',4)$ are equivalent to factorisations of $c$ through $k$, that is, to pairs $(\beta,\ c = k\,;\,\beta)$ (`TripleMaps.Key.key`).

In fibrewise terms, a null-homotopy is a family of maps $F_x : (k\,x = \mathrm{pt}) \to (c\,x = \mathrm{pt})$ between "$K(B,3)$-torsors". The proof separates a linear part from a translation:

- whether $F_x$ is a translate of the map induced by a fixed $\beta$ is a *proposition*, by the connectivity of the fibres and the truncation of the target;
- so it suffices to check it over the base point, since the base is connected;
- over the base point, the restriction of $F$ is a pointed map $\Omega K(B,4) \to \Omega K(B',4)$, and it has a *unique* delooping $\beta$ (Wärn's delooping theorem, `DeloopMaps`, Corollary 11);
- the translations are exactly the homotopy component, which stays as data.

No point is chosen in any fibre other than the base, no representative of a class is chosen, and lifts are not assumed unique. Finally, $\mathrm{Map}_*(K(B,4), K(B',4)) \simeq \mathrm{Hom}(B, B')$. The composite of these equivalences is, by definition, the canonical map `FibMap.fm`.

## 4. The homotopy precategory of triples

Composition of morphisms of triples is defined intrinsically. The coefficients compose, and the homotopies paste:

$$K(a\,;\,a', 2)\,;\,k'' = K(a,2)\,;\,k'\,;\,K(b', 4) = k\,;\,K(b\,;\,b', 4).$$

The identity uses $K(1, m) = \mathrm{id}$. The canonical maps of fibres preserve composition and identities up to pointed homotopy, that is, in the homotopy classes $[\mathrm{Fib}k, \mathrm{Fib}k'']_*$ (`TripleHomotopy.Maps3.rmap-comp`, `Maps1.rmap-id`). Put $\hom(t, t') = \|\mathsf{Mor}(t, t')\|_0$.

**Theorem 2** (`TripleHomotopy.Cat`, with `PostnikovClassification`)**.** *Triples, with the sets* $\hom(t, t')$ *and the induced composition and identities, form a precategory. The functor to the homotopy precategory of pointed simply connected* 3*-types has two parts:*

- *on objects it is* realise*, an equivalence of types;*
- *on hom-sets it is the class of the canonical map, a bijection by Theorem 1.*

*It preserves composition and identities.*

In the terminology of Ahrens, Kapulkin and Shulman (*Univalent categories and the Rezk completion*, Definition 6.9), this is an *isomorphism of precategories*: a fully faithful functor whose object map is an equivalence of types. A precategory may have an arbitrary type of objects (here 2-types), while its hom-types are sets. Neither precategory is univalent in their sense, and no such claim is made. By their Lemma 6.16, under univalence an isomorphism of precategories yields an equality of the two precategories; this consequence is not formalised here. No explicitly declared inverse functor is needed for the criterion. Equality in $[\mathrm{Fib}k, \mathrm{Fib}k'']_*$ is all the homotopy precategory requires, so no coherent pointed path for the preservation of composition is constructed.

## 5. $\pi_3(S^2)$ on the triple side

Take $t_0 = (\mathbb{Z}, \mathbb{Z}, \kappa)$, with $\kappa$ the cup square, and $P = \mathrm{Fib}\kappa$.

**Proposition 3** (`Pi3TripleUniversal`)**.** *For every triple* $t' = (A', B', k')$*, the map* $\hom(t_0, t') \to \mathrm{Hom}(\mathbb{Z}, A')$*,* $[a, b, H] \mapsto a$*, is a bijection.*

The proof is a direct computation on Mor:

- (H4), the bilinear classification of $[K(\mathbb{Z}, 2), K(B', 4)]_*$, shows that for each $a$ there is exactly one $b$ for which $K(a, 2)\,;k'$ and $\kappa\,;K(b, 4)$ are homotopic;
- (H3), $[K(\mathbb{Z}, 2), K(B', 3)]_* = 0$, shows that the homotopies between them form a single component.

The second point needs loops at an arbitrary point of $\mathrm{Map}_*(K(\mathbb{Z}, 2), K(B', 4))$. Writing $K(B', 4) \simeq \Omega K(B', 5)$, pointwise multiplication by $g^{-1}$ translates them to loops at 0 (`TripleComponents`).

On the sphere side, maps $S^2 \to \mathrm{Fib}k'$ correspond to maps $S^2 \to K(A', 2)$, hence to $\mathrm{Hom}(\mathbb{Z}, A')$. The marked class $s : [S^2, P]_*$ makes the triangle commute, because the canonical map of fibres lies over $K(a, 2)$. Therefore $s^* : [P, \mathrm{Fib}k']_* \to [S^2, \mathrm{Fib}k']_*$ is a bijection for every triple. By the classification of objects, this is the universal property for every pointed simply connected 3-type, and $\pi_3(S^2) \cong \mathbb{Z}$ follows (`Pi3TripleUniversal.Final.pi3-S2`, under the $S^2$ axiom only).

## 6. $\pi_4(S^3)$ on the triple side

*Looping* sends a pointed 2-connected 4-type $Y$ to the pointed simply connected 3-type $\Omega Y$; it is an embedding $\mathsf{SC4} \hookrightarrow \mathsf{SC3}$. Its image consists of the *deloopable* triples. On maps, looping is a bijection $[X, Y]_* \simeq [\Omega X, \Omega Y]_*$. The candidate is $\bar{t} = (\mathbb{Z}, \mathbb{Z}/2, \bar{k})$, with $\bar{k}$ the cup square reduced mod 2, and $R = \mathrm{Fib}\bar{k} = \Omega BR$.

**Proposition 4** (`Pi4TripleUniversal`)**.** *For every deloopable triple* $t'$*, the map* $\hom(\bar{t}, t') \to \mathrm{Hom}(\mathbb{Z}, A')$ *is a bijection.*

Again the computation is on Mor:

- (H4) and Statement T (deloopability forces $2h(1) = 0$) show that the required class factors through $\mathbb{Z}/2$;
- injectivity of $g \mapsto [\bar{k}\,;K(g, 4)]$ fixes $b$;
- (H3) controls the homotopies, as in Section 5.

On the sphere side, the source is $\Omega S^3$. Lifting there needs an existence hypothesis, and it is obtained from the surjectivity in the proposition above, so it does not use the earlier specialised lifting lemma at $R$. The looped marked class then has the universal property against every $\Omega Y$, and by the looping embedding statement M holds for every pointed 2-connected 4-type. Both instances used in the final step come from it: $BR$ itself and $\|S^3\|_4$. This gives $\pi_4(S^3) \cong \pi_4(BR) \cong \mathbb{Z}/2$ (`Pi4TripleUniversal.Final4.pi4-S3`, under the $S^3$ axiom only).

## 7. What is reused, and what is independent

The new sphere proofs reuse substantial mathematics from the earlier development:

- the cup square and its normalisation;
- (H3) and (H4);
- the cross-term and two-torsion analysis (Statement T);
- the delooping embedding and the looped form of statement M;
- the sphere-side computations at $S^2$, $S^3$ and $\Omega S^3$, together with the marked classes.

They do *not* use the earlier specialised mapping theorems as the reason the universal property holds. This was checked in a scratch copy. There the old results were deleted: `MapFib.Final.rep-fib`, the lifting lemma at $P$, the lifting at $R$ and $\Omega S^3$ through `MarkDirect.ExR`, `MarkDirect`'s marked theorem, `DeloopEmbedding.Mark.marked`, `FibreH3.fib-h3`, the presentation-based endpoints, and `Pi3Classified`/`Pi4Direct`. The copy still checks with all the new modules. The earlier endpoints remain unchanged in the development, as references.

## 8. Status and remaining qualifications

Everything above is checked. The whole development checks in a fresh process: 5633 declarations, 0 failed. All 32 `.opens` specifications pass the uniform checker. Two qualifications remain. Neither is a missing step in the checked sphere arguments.

- The theorem shows that the canonical map lies over $K(a, 2)$, and that it restricts on the base fibre to $\Omega K(b, 4)$. That the coefficient $b$ is the map induced on $\pi_3$, through the identification $\pi_3(\mathrm{Fib}k) \cong B$, is not formalised; it would make the interpretation of the classification more explicit.
- The six new `.opens` specifications were generated from the checker and audited against the informal proof *by kind*. Passing generated specifications is not the entry-by-entry unfolding audit, which remains to be done.